

Retail Data Engineering Project

SQL Question Bank (Master)

DE-009 — SQL JOINS

Ticket 1 — Customer Order Report

Business Requirement

Business Requirement

The Customer Success team wants a report of all customers along with any orders they have placed. Customers who have never placed an order should still appear.

Return

customer_id
customer_name
city
order_id
product_name
order_amount

Ticket 2 — Order Fulfillment Report

Business Requirement

The Fulfillment team only wants orders that belong to valid customers.

Return

order_id
customer_name
city
product_name
order_status

Ticket 3 — Customer Purchase Summary

Business Requirement

Marketing wants a report showing every customer and the products they have purchased. Customers without purchases should also appear.

Return

customer_name
city
product_name

Ticket 4 — Data Quality Check

Business Requirement

The Data Quality team wants to identify orders that reference customers that do not exist.

Return

order_id
customer_id
order_amount

Business Rules

- This is a real data engineering scenario.

Ticket 5 — Sales Analytics

Business Requirement

Finance wants to analyze revenue only for orders that belong to valid customers.

Return

customer_name
category
order_amount

Ticket 6 — Customer Activation Campaign

Business Requirement

The Marketing team wants to identify customers who signed up but have never placed a single order.

Return

customer_id
customer_name
city
signup_date

Business Rules

- Include only customers who have never placed any order.
- Each customer should appear only once.
- Sort by signup_date (oldest first), then customer_id.

Ticket 7 — Customer & Order Reconciliation Report

Business Requirement

The Data Governance team wants a reconciliation report between customer and order data.

Return

customer_id
customer_name
order_id
order_amount

Business Rules

- Include all customers, even if they have never placed an order.
- Include all orders, even if the customer record is missing.
- If information is unavailable on one side, it should appear as NULL.
- Sort by customer_id, then order_id.

Ticket 8 — Customer & Order Audit Report

Business Requirement

The Data Audit team wants a single report to verify the relationship between customers and orders.

Return

customer_id
customer_name
order_id
order_amount

Business Rules

- Include all records from both datasets.
- Missing information from either side should appear as NULL.
- Sort by customer_id, then order_id.

Ticket 9 — Regional Customer Campaign**Business Requirement**

The Marketing team has prepared two customer lists: Customers from Pune and Customers from Mumbai. They want one final customer list.

Return

customer_id
customer_name
city

Business Rules

- Include customers from Pune.
- Include customers from Mumbai.
- Sort by customer_id.

Ticket 10 — VIP Customer Lists**Business Requirement**

The Marketing team received two customer lists: List A contains customers from Pune. List B contains customers from Pune and Mumbai. They want one final customer list.

Return

customer_id
customer_name
city

Business Rules

- Combine both lists into one report.
- Sort by customer_id.